

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

Compound

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Half & Double
some formula

Just the way
you
use it

Half

Half

$$\sin(2A) = \sin A \cos A + \cos A \sin A$$

$$= 2[\sin A \cos A]$$

$$\cos(2A) = \cos A \cos A - \sin A \sin A$$

$$= \cos^2 A - \sin^2 A$$

$$(1 - \sin^2 A) - \sin^2 A$$

$$= 1 - 2\sin^2 A$$

$$\cos^2 A - (1 - \cos^2 A)$$

$$= 2\cos^2 A - 1$$

$$\tan(2A) = \frac{\tan A + \tan A}{1 - \tan A \tan A}$$

$$= \frac{2 \tan A}{1 - \tan^2 A}$$

θ in degrees

0°

30°

45°

60°

90°

180°

270°

360°

θ in radians

0

$\frac{\pi}{6}$

$\frac{\pi}{4}$

$\frac{\pi}{3}$

$\frac{\pi}{2}$

π

$\frac{3\pi}{2}$

2π

$\sin \theta$

0

$\frac{1}{2}$

$\frac{1}{\sqrt{2}}$

$\frac{\sqrt{3}}{2}$

1

0

-1

0

$\cos \theta$

1

$\frac{\sqrt{3}}{2}$

$\frac{1}{\sqrt{2}}$

$\frac{1}{2}$

0

-1

0

1

$\tan \theta$

0

$\frac{1}{\sqrt{3}}$

1

$\sqrt{3}$

undefined

0

undefined

0

